Presenter notes

Conceptual replication of Nie, Zheng and Sarkar (2022), “Competing with the Sharing Economy: Incumbents’ Reaction on Review Manipulation”, MIS Quarterly 46(3). One block per slide: what to say, the numbers, why we did it that way, the link to the paper, and the question most likely to come. Paper figures are the paper’s Texas results unless stated. ★ marks the sentence to say if there is time for only one.

The story in nine beats (memorise this, everything else hangs on it)

1. **Claim.** When Airbnb appears near a hotel, the hotel writes more fake praise about itself and attacks its hotel neighbours less, and high-end hotels react most.
2. **Measurement trick.** Fake reviews can only be posted on an open site (TripAdvisor), not on a verified one (Expedia), so the gap between the two is the manipulation.
3. **Identification trick.** Airbnb hosts choose where to list, so the paper instruments a hotel’s Airbnb with its competitors’ Airbnb, and tests the instrument on hotels never exposed.
4. **Our problem.** Three of six data sources are unobtainable, so we rebuilt the design twice: Texas with a stand-in measure, Europe with the real measure.
5. **Texas.** Every sign reproduces, the headline coefficient lands on the paper’s number, the instrument is strong, the placebo and parallel-trend tests pass; precision is lower.
6. **Power.** Where the test has power, pairs, triplets, text and synthetic control, the result is significant.
7. **Europe.** Nothing can be identified because every hotel is treated; the design has a domain the paper does not state.
8. **The one claim that fails.** High-end versus low-end rests on proprietary tiers; with ours it is not significant and reverses in the group tests.
9. **Extensions.** The effect lives in Airbnb’s growth phase (Texas after 2016 looks like Europe), a text classifier gives the same 0.02, and Europe validates the Texas proxy ($r = 0.69$).

Cover

SLIDE 1

This is a conceptual replication of Nie, Zheng and Sarkar, MISQ 2022. Same question, same design, public data in place of their proprietary sources, plus three extensions. Before the slides, two minutes of background.

The paper asks what happens to hotels' review manipulation when Airbnb arrives. Hotels manipulate in two directions: fake five-star reviews of themselves (self-promotion) and fake one-star reviews of the hotel next door (demotion). Airbnb takes guests but cannot be attacked on a hotel review page. The theory: demotion is a tit-for-tat game within a strategic group of same-tier neighbours; when guests can leave for Airbnb, winning that fight is worth less, so demotion falls, while self-promotion still works against any competitor, so it rises. High-end hotels, hurt most by Airbnb on price, should react most. That is H1, H2 and H3.

You cannot see manipulation directly. Mayzlin, Dover and Chevalier (2014) noticed that TripAdvisor lets anyone review and Expedia only lets people who stayed, so fakes can only land on TripAdvisor. The five-star share on TripAdvisor minus the five-star share on Expedia is self-promotion; the same gap for one- and two-star reviews is demotion received. Demoting others is inferred by spreading each competitor's received demotions evenly over that competitor's competitors.

Airbnb hosts choose where to list, so Airbnb supply is endogenous. The paper instruments a hotel's Airbnb with Airbnb near its competitors, excluding its own, and shows the instrument has no direct effect in hotels that never had an Airbnb. ★ Hold these three ideas, the claim, the measurement gap and the competitors' instrument, and every table that follows is one of them being tested.

IF ASKED why conceptual rather than direct: no data or code package exists for the paper; their TripAdvisor and Expedia crawls, STR tiers and AirDNA prices cannot be obtained. The design can be rebuilt; the data cannot be copied.

Replication strategy, two independent setups

SLIDE 2

Because the data cannot be copied, the replication is a reconstruction, and we did it twice on purpose. Texas keeps the paper's geography and design: Austin, Dallas and Fort Worth, 448 hotels, 2011 Q1 to 2019 Q2, 10,068 hotel-quarters. Its weakness is that no verified-review site is public for Texas, so manipulation is measured with a stand-in, reviews from one-time reviewer accounts. Europe, six cities, 903 hotels, 2015 Q3 to 2017 Q3, 7,101 hotel-quarters, has the paper's exact measure, TripAdvisor against Booking.com where only guests who stayed can review, but a different geography.

Each panel is a different test of the same claim. If Texas shows the effect, the result holds with a different measure. If Europe shows it, it holds with a different geography. ★ As it turned out, Texas shows it and Europe cannot be tested at all, and the reason it cannot is itself a finding, which is slide 14.

Both panels use the paper's unit (hotel-quarter), radius (1 km), lag (previous quarter), fixed effects (hotel, quarter, city × season) and estimator (2SLS, errors clustered by hotel). Our Texas horizon is longer than the paper's 2008 to 2015 because that is what the review corpus covers; that turns into Extension 1.

Bridge: "Here is what each of their six sources became."

Data and what each source became

SLIDE 3

Walk the table. TripAdvisor crawl → HotelRec, a public corpus of 50 million TripAdvisor hotel reviews. Expedia → nothing public exists for Texas, unobtainable; Booking.com's 515K review set is the European substitute, with the same verified-stay property. Airbnb.com and AirDNA → Inside Airbnb snapshots, with AirDNA's realised prices replaced by listed prices. Smith Travel Research, which gave the paper its price tiers and room counts → commercial; replaced by a list of economy brands (Motel 6, Super 8, Days Inn, Econo Lodge and so on), 26 percent of hotels against the paper's 24. Texas Comptroller tax filings for operating periods → the first-to-last review window. Expedia coordinates → read from each hotel's own TripAdvisor page.

★ One input has no substitute at all for Texas, the verified-review benchmark, and that is why the Texas measure is a proxy. The footnote on Tables 6 and 7 is there because they depend on the STR covariates.

IF ASKED whether the substitutes bias the result: the brand list is coarser than STR tiers, which matters for H3 and only H3. The other substitutes change levels, not the within-hotel variation the fixed effects use.

Data pipeline

SLIDE 4

Four steps. Filter: one resumable pass over the 50 GB file keeps 2.54 million reviews for 10,732 hotels in the nine cities. Link: Booking hotels matched to TripAdvisor by name, 1,125 of 1,492 at similarity 90 or better. Every distance in the design depends on coordinates, and the review file has none; a name-based geocoder put about 30 percent of chain hotels at the wrong branch, so all 449 Texas hotels were placed from the coordinates embedded in their own TripAdvisor pages. Airbnb: 253 thousand listings, entry quarter from host tenure, listings within 1 km counted per hotel-quarter and lagged. Panels: 10,068 and 7,101 hotel-quarters.

The hotel counts: Texas 449 geocoded, one Econo Lodge has no review after 2008, so 448 in the panel. Europe 1,492 → 1,125 matched → 939 with coordinates → 903 with at least one quarter that has both TripAdvisor and Booking reviews. In both, a hotel counts if it has one usable hotel-quarter, which is how the paper counts its 2,188.

★ Python builds the panels because the geography is set arithmetic (listings within 1 km, unions of competitors' listing sets, triplets); every regression, test and figure is Stata, and the two agree to four decimals.

Measures

SLIDE 5

Construct by construct. Self-promotion: the paper uses the TripAdvisor minus Expedia five-star gap. Texas: the share of a hotel-quarter's reviews that are five-star and written by a one-time reviewer, an account with a single review in the whole Texas corpus (Luca and Zervas 2016: fake accounts post once). Europe: the paper's gap, TripAdvisor five-star share minus the Booking share of scores 9 to 10 on Booking's 0 to 10 scale. Demotion received: the mirror image. Demoting others: built exactly as in the paper from whichever demotion measure the panel has. Low-end: STR tier in the paper, the brand list here; in Europe the Booking sample is upscale and has two budget hotels, so Model 3 is not estimable there and Europe tests H1 and H2 only. Airbnb supply and the instrument are built identically to the paper.

★ The question you will get is "why should one-time reviewers measure manipulation?" The answer is slide 30: in Europe, where both measures exist, the one-time-reviewer excess correlates 0.69 with the paper's own gap, and slide 29 gives the same coefficient from review text. Say you will come back to it.

IF ASKED about the lag: Airbnb at t−1 counts listings whose host had started by the end of the previous quarter. Inside Airbnb no longer publishes host_since, so start = scrape date minus tenure in years and months, capped at January 2011; the paper itself reports almost no supply before 2011 Q4.

Bridge: "Now the tables, in the paper's order. Every regression table has the same three columns per outcome and the same bottom rows, so let me say once how to read them."

Table 1, summary statistics

SLIDE 6

How to read every table that follows: Model 1 has Airbnb and competing hotels only; Model 2 adds the hotel's own review count and rating shares; Model 3 adds the low-end interaction, so the Airbnb row becomes the high-end effect. The top number is the coefficient, the bracket is the standard error, stars are p below 0.05, 0.01, 0.001. All treatments are in logs, so doubling Airbnb moves the outcome by 0.69 times the coefficient. Kleibergen-Paap F is the strength of the instrument; above 10 is fine, the paper has 796, we have 86.

★ The panel is a smaller image of the paper's: 12.4 TripAdvisor reviews per hotel-quarter against 10.9, 3.3 competing hotels against 5.0, 26 percent low-end against 24. The one number that looks off, 11.8 competing listings against 3.3, is the horizon; in 2015 Q4 we have 9.29 and the paper 9.19.

```
Hotels 2,188 vs 448 | hotel-quarters 32,122 vs 10,068 | reviews/qtr 10.9 vs 12.4 |
Airbnb 2015Q4 9.19 vs 9.29 | Europe: 903 hotels, 526 listings per hotel
```

Table 2, self-promotion (H1)

SLIDE 7

The headline. The paper finds 0.021 with three stars: doubling Airbnb raises the suspicious five-star share by about 1.5 points. We find 0.022, the same number, with a standard error three times the paper's, so the interval includes zero. ★ Directionally reproduced, underpowered, and that phrase recurs. Three reasons the error is larger: a third of the observations, three cities instead of 64, and a proxy outcome, which adds noise to the dependent variable and inflates the standard error without biasing the coefficient. The instrument is strong in both, F of 86 against 796, so this is a precision problem, not an identification problem. The low-end interaction has the paper's negative sign (low-end hotels respond less) but is a third of its size and not significant: the first appearance of the heterogeneity problem.

```
Paper  0.021*** (0.006) | 0.020** | 0.020*** | x Low-end −0.054* (0.022) | N 32,122 | F
796
Ours   0.022 (0.017)    | 0.018   | 0.022    | x Low-end −0.023 (0.023)  | N 10,068 | F
86
```

VALUE OF THE RESULT: the point estimate is reproduced from independent data with an independent measure; what is missing is precision, which the pair, triplet, text and synthetic-control tests supply later.

Table 3, demoting others (H2)

SLIDE 8

H2 is the strategic-group prediction: hotels attack each other less once Airbnb takes guests. The paper finds −0.035 on Airbnb and −0.297 on competing hotels, both three stars. We find −0.128 and −0.157 in 2SLS, both the right sign, neither significant, and the competing-hotels coefficient is −0.260 with two stars in plain OLS with fixed effects. ★ The competing-hotels result is the purest form of the theory: the more incumbents in your group, the less each one demotes, and that reproduces at conventional significance. The Airbnb effect on demotion is the weaker of the two hypotheses at the hotel level, because demoting others is inferred twice over, first the one-time-reviewer negative share, then spread over competitors. It is better tested where the attacks are observed directly, which is the next slide.

```
Paper  −0.035*** (0.009) | CompHotels −0.297*** (0.044) | N 24,713
Ours   −0.128 (0.090)    | CompHotels −0.157 (0.099), OLS −0.260** | N 7,487 | F 70
```

Table 4, mutual demotion at the pair level (H2)

SLIDE 9

For a pair of competitors A and B, the sum of demotions in both directions is observed without the attribution inference, and pairs multiply the sample. Pair Airbnb is the intersection of the two listing sets, competing hotels the intersection of competitor sets, the instrument is rebuilt from shared competitors, with pair, quarter and city × season fixed effects, clustered by pair. ★ This is the first Texas result to reach significance: −0.191 with a star, against the paper's −0.107 with three. The one thing that does not reproduce is the low-end interaction, +0.136 in the paper (budget hotels keep fighting), −0.072 and not significant here. Flag it; it returns in C1 and in the verdict.

```
Paper  −0.107*** (0.008) | −0.092*** | −0.092*** | x Low-end +0.136*** | N 104,424
pair-qtrs
Ours   −0.191* (0.079)   | −0.177*   | −0.164*   | x Low-end −0.072 (n.s.) | 991 pairs,
N 15,516 | F 106
```

WHY PAIRS MATTER: power and measurement at once. The outcome is a count of attributed attacks between two named hotels, and 15,516 pair-quarters against 7,487 hotel-quarters.

Table 8, being demoted

SLIDE 10

The paper flips demotion around and asks whether a hotel receives fewer attacks when Airbnb grows near it, and finds the opposite, 0.123 with a star: more Airbnb, more demotion received. Its explanation is a composition effect. The hotel with more Airbnb attacks less, its neighbour with less Airbnb keeps attacking, so the first hotel ends up relatively more demoted. We find 0.022, right sign, p about 0.09, smaller because our outcome is a share of one-time-reviewer negatives rather than a scaled count, so only the sign is comparable. ★ Table 8 is the mirror of Table 3 and both move the way the paper says.

```
Paper 0.123* (0.048) | 0.137** | 0.139** | N 32,122     Ours 0.022 (0.013) | 0.021 |
0.017 | N 10,068
```

Table 5, validity of the instrument

SLIDE 11

Explain the logic first. The instrument is Airbnb near your competitors. If it worked through any channel other than your own Airbnb exposure, it would move manipulation even in hotels that never had an Airbnb within 1 km. So regress manipulation directly on the instrument in two groups. Unexposed hotels: nothing should happen. Exposed hotels: something should. Paper: 0.004 and −0.009, nothing, against 0.012 and −0.032, two and three stars. Ours: −0.021 and 0.286, nothing, against 0.005 and −0.097 with a star. ★ The placebo pattern reproduces. This test exists only in Texas, where 98 of 448 hotels never get an Airbnb competitor; in Europe no such hotel exists. Columns 5 to 8 of the paper repeat this on a PSM-matched sample and need STR covariates.

```
Unexposed: paper 0.004 (0.019), −0.009 (0.015) | ours −0.021 (0.023), 0.286 (0.172)
Exposed:   paper 0.012** (0.004), −0.032*** (0.010) | ours 0.005 (0.007), −0.097*
(0.045)
```

IF ASKED what would violate exclusion: a neighbourhood shock that raises both competitors' Airbnb and your manipulation directly, say a new convention centre. The fixed effects absorb the slow-moving and seasonal versions; this placebo checks the rest.

Table 6, look-ahead PSM

SLIDE 12

Hotels that get Airbnb early may differ from those that get it late. The instrument handles that in principle; matching handles it by construction. Figure 1 of the paper defines Group A (Airbnb in Period 1), B (only in Period 2), C (never). Look-ahead matching pairs each A hotel with a B or C hotel that looks like it before treatment. Ours: A 296, B 55, C 98. The paper matches on STR price, rooms, chain status and location type, none of which we have, so we match on pre-period review volume, mean rating, five-star and one-to-two-star shares, tier, competing hotels, city and opening year: nearest neighbour without replacement, caliper 0.1, 101 pairs, mean bias 30 to 12 percent, joint balance test no longer rejecting. Then the same 2SLS on the matched sample. ★ Self-promotion 0.050 and 0.035, demoting others −0.52: the paper's signs (0.034*, −0.084**) without significance on a fifth of the sample. Model 3 has no first stage in the matched sample because almost no matched hotel is low-end.

```
Paper 0.034* (0.014) | 0.031* | demote −0.084** (0.028) | N 10,759 / 8,630
Ours  0.050 (0.042)  | 0.035  | demote −0.515 (0.408)  | N 4,424 / 2,916 | F 13.5
```

Table 7, generalized synthetic control

SLIDE 13

The paper's third design, so the reader sees the result does not hinge on the instrument. Xu (2017): fit a factor model to the untreated hotels, project what each treated hotel would have done without Airbnb, take the gap. Treatment is binary, the first competing listing within 1 km. The estimator exists only in R, in the authors' own package fect; cross-validation chose the number of factors; it accepts missing outcomes so we used all 448 hotels. ★ Self-promotion rises by 0.033 (p = 0.03) for all hotels and 0.039 (p = 0.02) for high-end, against the paper's 0.029 and 0.033, nothing for low-end in either. The plot: effects flat at zero before treatment, positive after. Demoting others is null here where the paper has −0.037 for high-end. The lower table is Stata's sdid, which needs a balanced panel of 88 hotels, as a cross-check.

```
Paper 0.029* all | 0.033* high | −0.003 low | demote −0.034, −0.037*, 0.192
fect  0.033 (0.015) | 0.039 (0.017) | 0.009 (0.042) | demote 0.046, 0.004, 0.417 (all
n.s.) | 165 treated, 101 controls
```

IF ASKED why r = 0: after hotel and quarter fixed effects there is little common time variation in a review share, so extra factors only add prediction error; the estimator is the paper's, applied as specified.

Europe, the paper's measure, no identification

SLIDE 14

Transition: "Texas tested the finding on its home turf with a proxy. Europe has the paper's own measure." Then the surprise. The coefficients mean nothing: first-stage F of 2.8 to 5.0 against the Stock-Yogo value of 16.4 and the paper's 796; pair-level F of 0.3. The reason is mechanical. Every European hotel has hundreds of listings within 1 km, so no hotel is untreated. Hotel fixed effects absorb each hotel's level, quarter fixed effects absorb the city-wide growth, and what remains of competitors' Airbnb is noise. ★ This is not a failed replication; it is a boundary condition. The identification strategy works where Airbnb is sparse and uneven, mid-2010s Texas, and fails by construction where it is dense, which by now includes most large cities. The paper does not state this limit. The stricter version: Table 5's placebo group does not exist here, so exclusion cannot even be checked.

```
Europe 2SLS: promote −0.162 (0.092), demote −0.563 (0.503), being demoted −0.002
(0.059) | F 2.8 / 5.0 / 2.8 | N 7,101
```

Figures A1 and A2, model-free evidence

SLIDE 15

The paper's Appendix A pictures, redrawn in Stata from our data. Model-free means no instrument, no regression, just the averages. A1: average TripAdvisor ratings rise quarter by quarter as the Airbnb line climbs. A2: high-end hotels show a rising positive share with a flat negative share, low-end hotels show neither. ★ This is the raw pattern the hypotheses need, ratings improving exactly as Airbnb grows and on the positive side only, and it reproduces cleanly over a longer window. Everything before this slide asked whether it is causal; this slide shows the correlation it started from.

Reproduction status, table by table

SLIDE 16

The instructor's second and third requirements on one slide. Three bands. Reproduced in direction and magnitude: Tables 1, 2, 3, 8, 5, 6, G1, B1, F1, H1, E1 and the figures. Reproduced with significance: Table 4, Table C1, Table 7 for self-promotion, and the extensions. Not reproduced: the low-end interaction wherever it appears, the demotion side of Table 7, and Europe as a whole, which is not identified. ★ Say what "reproduced" means: the table was rebuilt with the same specification and the signs and magnitudes compare, not that the numbers coincide. The obstacle column says what changed each time.

Results reported in the text

SLIDE 17

Numbers that live in sentences, not tables. The first-stage F, 796 in the paper, 86 here. ★ The effect-size calculation: Airbnb grew from 0.55 to 9.19 listings per hotel between 2011 and 2015, which through 0.021 implies a 5.6-point rise in self-promotion; ours grew from 0.79 to 9.29 and implies 5.4 points, the strongest sign that the two panels describe the same market. Hotels with an Airbnb competitor, 48 percent in the paper, 76 here because the horizon runs to 2019. Mean ratings by tier, 3.0 and 4.0 against 3.23 and 4.13, which justifies the different thresholds for low-end hotels. The Appendix D and G statistics. The one in-text result that cannot be obtained is the management-response rate, which lives on Expedia.

Bridge: "The paper has nine appendices, most asking whether a modelling choice drives the result. We ran all of them, in the order of what they test."

Table G1, Airbnb segments by price

SLIDE 18

Should a luxury hotel count a $30 room as a competitor? Listings below the 24th percentile of price per person ($25 in the paper from AirDNA, $34 here from listed prices) are matched to low-end hotels, the rest to high-end, and the instrument is rebuilt the same way. ★ Paper 0.020, 0.019, 0.018, unchanged from Table 2; ours 0.026, 0.021, 0.022, unchanged from our Table 2. Segmenting changes nothing in either.

```
Paper 0.020** | 0.019** | 0.018** | demote −0.034***      Ours 0.026 (0.016) | 0.021 |
0.022 | demote −0.126 (0.079) | F 96
```

Table B1, parallel trends (relative-time model)

SLIDE 19

The identification check an econometrician asks for first. The design is a staggered difference-in-differences with a continuous treatment, so the paper dichotomises: Airbnb_Binary = 1 once any listing is within 1 km, then dummies for j = −2, −1, 0, +1, +2, +3 quarters around the first treated quarter (Autor 2003), with hotel and quarter fixed effects. If the leads at −2 and −1 are significant, hotels were changing before Airbnb arrived and the story is suspect. Paper: leads insignificant, treat(0) 0.060* for self-promotion and −0.108* for demoting others. Ours: 351 hotels become treated, 98 never do; leads −0.005 and 0.011, nothing; self-promotion 0.039* one quarter after treatment; demoting others never moves. ★ No pre-trend, and the response arrives one quarter after the listing is first counted, which is what a lagged treatment should produce. The figure shows it at a glance: dots on zero before, a positive dot after.

```
Self-promotion, ours: −0.005 (0.017), 0.011 (0.018), 0.011 (0.016), 0.039* (0.016),
0.004, 0.003
Paper: −0.079 (0.234), −0.035 (0.025), 0.060* (0.026), −0.046, 0.010, 0.040
```

IF ASKED why never-treated hotels matter: with all six dummies and a hotel fixed effect, the dummies sum to one for every treated hotel inside the window, so the 98 never-treated hotels and the quarters outside the window identify the levels.

Table F1, uniform thresholds

SLIDE 20

The paper counts four and five stars as promotion for low-end hotels because their ratings are lower on average. A critic could call the result an artefact of that choice. Appendix F uses five stars and one-to-two stars for everyone, Mayzlin's original definitions. ★ Paper 0.027, 0.025, 0.026 and −0.036, all significant; ours 0.024, 0.020, 0.019 and −0.16, identical to our Tables 2 and 3. Thresholds set the level of the measure, not how it responds to Airbnb.

Table H1, all neighbouring hotels as competitors

SLIDE 21

The main tables assume a budget motel does not attack the Hilton next door, so competitors are same-tier only. Appendix H counts every hotel within 1 km, which changes the competitor count, the demoting-others attribution and the instrument. Paper 0.022, 0.020, 0.021 and −0.018, essentially unchanged. Ours 0.011, 0.010, 0.014 and −0.076, all the right sign, self-promotion about half the main-table size, competing hotels −0.208 with two stars, first stage above 100. ★ Cross-tier competitors dilute the instrument with Airbnb near hotels that do not compete for the same guest, which is a reason to prefer the same-tier definition, as the paper does.

Table C1, hotel triplets

SLIDE 22

Table 4 with three hotels that all compete with each other, a stricter test since all three pairwise distances must be under 1 km. Mutual demotion is the sum of the six directed demotions; Airbnb and competing hotels are intersections of the three sets; the instrument comes from shared competitors; 2,403 triplets, 24,966 triplet-quarters. ★ log(Airbnb) is −0.423 with three stars, against the paper's −0.259, the strongest estimate in the whole replication, and it is H2, the hypothesis that looked weakest at the hotel level. The lesson: the demotion effect lives at the group level, where measurement is cleaner and the sample larger, in the paper and here. The low-end interaction is −0.127** here and +1.667* in the paper, the same reversal as Table 4.

```
Paper −0.259*** (0.011) | −0.185*** | −0.178*** | x Low-end +1.667* | N 437,695
Ours  −0.423*** (0.108) | −0.386*** | −0.347**  | x Low-end −0.127** | N 24,966 | F 127
```

Table E1, Gaussian kernel

SLIDE 23

The 1 km radius is a wall: a listing at 999 metres counts fully, at 1,001 not at all. Appendix E replaces it with $\exp(-d^2/2h^2)$ at h = 1 km, so a listing at the door contributes 1, at 1 km 0.61, at 2 km 0.14, and demoting reviews are attributed by the same weights. The paper reports 1 km and 0.5 km as consistent with its main results but does not tabulate them. Ours: self-promotion 0.013 to 0.017, positive, not significant; demoting others +0.05 at 1 km and −0.26 at 0.5 km, neither significant; competing hotels negative and significant; first stage in the hundreds because the kernel instrument has more variation. ★ This is our weakest robustness check and I say so. The hard cutoff gives the cleaner picture in our data; in Europe the kernel would have to be the main specification, since 1 km holds a whole district there.

Figure D1, trip types, Europe

SLIDE 24

The rival explanation: maybe Airbnb takes the price-sensitive guests, the remaining guests are happier, and ratings rise without any cheating. The paper shows the mix of trip types is stable over time (repeated-measures ANOVA p near 1, KPSS stationary). TripAdvisor has no trip-type field, but Booking.com tags every review, so this is reproducible in Europe only. ★ Friedman test across quarters p = 1.00, KPSS p-values 0.07 to 0.10, every share stationary; the composition story is ruled out here as in the paper.

Obstacles

SLIDE 25

The two a reader would not anticipate. Host registration dates: the paper dates each listing by when the host joined, Inside Airbnb no longer publishes that field, so we rebuilt it from tenure in years and months relative to the scrape date, capped at January 2011. Coordinates: no review file has them; a name-based geocoder put 30 percent of hotels at the wrong branch of the same chain, which would have corrupted every distance, so all 449 were placed from their own TripAdvisor pages. The rest in a line each: Booking has no hotel id, so name matching; no STR tiers, so a brand list; only listings alive at crawl time are observed, which the paper shares and which attenuates the Airbnb effect; no tax filings, so the operating period is the first-to-last review window. ★ Every obstacle is logged with what replaced it, so the assessment is explicit about what changed.

Overall reproducibility

SLIDE 26

The instructor's first requirement, and the slide to remember. Texas: directionally reproduced, underpowered; every main coefficient has the paper's sign, the self-promotion estimate matches to the third decimal, the instrument is strong, the placebo pattern of Table 5 holds and the parallel-trend test passes; significance arrives where the test has power. Europe: the paper's exact measure can be built, but the design does not identify in markets where every hotel is treated, a boundary condition the paper does not state. Heterogeneity: the high-end versus low-end contrast depends on proprietary STR tiers and is the least robust claim; with brand tiers the self-promotion interaction has the right sign but is not significant, and the pair and triplet interactions reverse. ★ If there is one slide to remember, it is this one.

What the replication puts in question

SLIDE 27

None of these overturn the paper; they narrow it. One, the domain of the instrument, which needs sparse, uneven supply and so dates the design to an era. Two, the 1 km radius, meaningful in Texas, meaningless in Paris. Three, the headline heterogeneity, which rests on tiers nobody outside STR can rebuild and is the one claim that changes sign in our data. Four, survivorship in Airbnb supply: both the paper and we date listings from a snapshot of listings alive at crawl time, churn is higher where density is higher, so the treatment is measured with error that attenuates the estimates, which makes our Texas numbers conservative. Five, the comparability of the two platforms, which Booking's trip-type tags let us test and which holds. ★ If asked what I would do as the authors: report the first stage by city, make the kernel the main specification in dense markets, and show the heterogeneity with a tier measure others can rebuild.

Extension 1, the effect belongs to Airbnb's growth phase

SLIDE 28

Transition: three extensions, each answering a question the replication raised. The paper's Appendix I splits at 2011 Q4 and finds nothing before the inflection point; our panel runs four years past the paper's, so we cut Texas into three periods and ran the same 2SLS in each. Before 2012 Airbnb was tiny, under one listing per hotel, no effect, matching the paper. 2012 to 2015, the paper's own window, Airbnb grew from 1 to 9 listings per hotel and self-promotion rises with it, 0.107 and significant. From 2016, 12 to 34 listings per hotel, the effect is gone. ★ The paper's result is real but belongs to the years when Airbnb was still spreading, and that is the same reason Europe shows nothing: Texas after 2016 is Europe. This slide turns the Europe non-result into a finding.

```
Self-promotion: ≤2011Q4 −0.061 (0.080) | 2012–15 0.107* (0.045) | ≥2016 0.040 (0.114)
Paper I1 (demoting others): before 0.010 (n.s.), after −0.025*
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IF ASKED why 0.107 is larger than the full-sample 0.022: the full sample averages a real effect with the zero of 2016 to 2019. Sub-period first stages are weaker (F 10 to 42).

Extension 2, a text-based manipulation measure

SLIDE 29

The paper never reads a review; it infers fakes from a gap between websites. We have 2.5 million texts, so we tried the direct route. Ott et al. (2011) built a corpus of 1,600 hotel reviews, half commissioned fakes, the standard training set for deception detection. A TF-IDF plus logistic-regression classifier reaches 88 percent five-fold accuracy against a published 89. We scored every review and took, per hotel-quarter, the share of five-star reviews flagged deceptive. ★ That share rises with Airbnb at 0.023 with a star, almost exactly the paper's 0.021, and significant; fake-looking negatives do not move; the text measure correlates 0.46 with the one-time-reviewer measure. Say the caveat out loud: the training reviews are about Chicago hotels in 2011, so place and brand words were removed so the classifier could not learn geography. Three measures built three ways give about 0.02.

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Fake 5-star share: 0.023* (0.010) | 0.022* | 0.022*   Fake 1-2-star share: −0.007
(0.007)   N 10,068 | F 86
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Extension 3, validating the proxy on a verified platform

SLIDE 30

This closes the loop opened on slide 5. Texas had no verified site, so we used one-time reviewers as a stand-in. Europe has both TripAdvisor and Booking, and Booking records each reviewer's review count. One-time reviewers write 46.9 percent of the five-star and one-to-two-star reviews on TripAdvisor but 27.1 percent on Booking, where you cannot post without a stay. ★ The gap between the two, hotel by hotel and quarter by quarter, correlates 0.69 with the paper's own measure, so the stand-in captures the same thing. As everywhere in Europe, the gap does not respond to Airbnb because the instrument is weak.

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One-time share: TripAdvisor 46.9% vs Booking 27.1% | r(excess, Mayzlin gap) = 0.69 |
2SLS on Airbnb −0.102 (0.062), weak IV
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Conclusion

SLIDE 31

Three sentences and stop. The paper's Texas result survives a conceptual replication in direction, in magnitude and in its placebo and parallel-trend logic, but not in statistical strength at the hotel level, and it reaches significance at the pair, triplet, text and synthetic-control levels. Its identification strategy has a domain, sparse and uneven Airbnb supply; outside it, in dense European cities or in Texas after 2016, the instrument has nothing to work with. The high-end versus low-end contrast is the claim most dependent on proprietary data and the least robust.

Code map if asked: 01 to 04 build the data; 05 and 06 the main tables in Python and Stata; 07 figures and in-text results; 08 pairs; 09 extensions; 10 the classifier; 11 appendices B, C, F, H; 12 Table 6, Table 7 (sdid) and Appendix E; 13 Table 7 in R (fect). Stata outputs in results/stata, comparison in results/comparison_with_paper.md.

Question bank

KEEP AT HAND

Why is Airbnb endogenous? Hosts list where demand is, which is where hotels are busy and may already be manipulating.

Why is competitors' Airbnb valid? Relevance: Airbnb grows in neighbourhoods (F = 86). Exclusion: a listing next to another hotel changes your behaviour only through the supply you face; Table 5 shows no direct effect where you were never exposed.

Why 2SLS and not DiD? No single event; treatment is a count growing at different times. Appendix B is the DiD-style timing check.

Why are your SEs three times larger? A third of the observations, three cities, and a proxy outcome that adds noise to the dependent variable.

Is the one-time-reviewer measure valid? r = 0.69 with the paper's gap in Europe; the text classifier gives the same coefficient.

Why did Europe fail? Identification, not the result: every hotel is treated, so after fixed effects the instrument has no variation (F 3 to 5).

Why is the low-end interaction different? STR price tiers versus a brand list; with ours it is not significant and reverses in the group tests. The claim we can least reproduce.

Why R for Table 7? GSC exists only in R (fect). sdid in Stata needs a balanced panel of 88 hotels, so it is a cross-check.

What is GSC doing? Factor model on untreated hotels, counterfactual for each treated hotel from its pre-treatment fit, average the post-treatment gap.

What is look-ahead matching? Early-treated hotels matched to not-yet-treated or never-treated hotels on pre-treatment covariates.

Why Python for the panels? Distance and set operations; every estimate and figure is Stata; the two agree to four decimals.

Biggest threat to the paper? Survivorship in the Airbnb snapshot (attenuation) and the proprietary tiers behind the heterogeneity claim.

What surprised you? Demotion is weakest at the hotel level and strongest at the triplet level (−0.42***); self-promotion vanishes after 2016, exactly the European pattern.